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UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

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# The study of Higgs properties in the four-lepton final state at CMS

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*Author:*

Tahir JAVAID

*Supervisor:*

Prof. Guoming CHEN

*Co-supervisor:*

Prof. Mingshui CHEN

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Chinese Academy of Sciences





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*Dedicated to my family who endured missed birthdays,  
missed festivals and countless days alone so that I may  
make history*



# Chapter 1

## Introduction

The Standard Model (SM) of the particle physics gives successful explanation about the constituents of the physical matter and the fundamental forces through which they interact with each other and with composite particles. The electromagnetic, strong and weak forces are described by dedicated theoretical descriptions Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD) and Quantum Flavordynamics (QFD) which are encoded in the SM. Later on, electromagnetic and weak forces among these, were unified as electroweak (EWK) force at the order of 100 GeV unification energy scale. Sheldon Glashow, Abdus Salam, and Steven Weinberg had significant contribution in this milestone. The experimental verifications of the existence of the electroweak interaction were established by the In 1973, existence of neutral currents announced by the Gargamelle collaboration and then in 1981, the discovery of two massive gauge bosons (W and Z) by UA1 and UA2 collaborations verified the existence of EWK force. These boson should be massless (like photon in QED) according to Goldstone's theorem. To overcome this issue, spontaneous symmetry breaking was introduced that generates the masses of the particles by the Higgs mechanism and separates weak interaction from electromagnetic (EM) interaction [**Reference7**, **Reference8**, **Reference9**, **Reference10**, **Reference11**, **Reference12**]. Higgs boson, a new scalar particle, was introduced by the theory being last missing patch of the SM. This triggered the search for this boson of the SM at experimental facilities which lasted for several years. One of the main design and construction motivations of the Large Hadron Collider (LHC) at the CERN was also to look for this new particle given unitary and above the Large Electron-Positron (LEP) constraints, i.e from 114 GeV to 1 TeV. After a long search, the CMS and the ATLAS collaborations of LHC, announced the existence of the heavy scalar boson (the Higgs) on 4 July 2012 [**Reference20**, **Reference21**]. Immediately after the discovery, further tests to measure its properties were begun. Mass of the newly discovered particle, the only free parameter in Higgs sector, has been measured to be  $m_H = 125.09 \pm 0.21(stat.) \pm 0.11(syst.)$  GeV [**Reference22**]. Other properties such as spin, parity [**Reference21a**] and couplings of this boson to different particles [**Reference21b**, **Reference21c**] have also been measured and observed to consistent with the predictions of SM. Therefore, the new particle very probable appears as the SM Higgs boson, affiliated with EWK spontaneous symmetry breaking. SM has passed enormous experimental tests in last 50 years and no significant deviation has been seen yet. Experimental evidence of the Higgs

boson is the biggest achievement of the SM. Other success includes the experimental verification of its prior predictions about the W and Z gauge bosons, gluon, top and charm quarks.

Although SM has passed numerous experimental tests with great precision, it is not considered to be theory of everything. There are many phenomena which are not explained by SM i.e Hierarchy problem, neutrino mass and existence of dark matter and dark energy etc. There are several theoretical extensions posed in SM such as the Minimal Super Symmetrical Model (MSSM) and the Next-to-Minimal Super Symmetric Standard Model (NMSSM) or completely new theoretical descriptions, such as string theory and extra dimensions. More precise Higgs boson properties measurements are needed to capture any possible deviation from SM predictions which can be a doorway to new physics.

Model independent production cross section for a particular process is an important property of H boson, where the cross section gives the possibility of the occurrence of that process. In this dissertation, measurements of the integrated or inclusive and differential fiducial cross sections for H boson production in  $H \rightarrow 4\ell$  decay channel ( $\ell = e, \mu$ ) in proton-proton collisions at  $\sqrt{s} = 13$  TeV are presented. These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of SM at the TeV energy scale. Any deviation of the integrated or the differential cross sections from SM expectations would provide a direct hint to BSM contributions. Nearly similar measurements have already been reported by the ATLAS experiment in  $H \rightarrow \gamma\gamma$  [ATLAS:Hgg\_8TeV, ATLAS:Hgg\_8TeV\_13TeV],  $H \rightarrow 4\ell$  [ATLAS:H4l\_8TeV, ATLAS:H4l\_8TeV\_13TeV] decay channels, combination of both channels [ATLAS:H4l\_Hgg\_8TeV, ATLAS:H4l\_Hgg\_8TeV\_13TeV] and by the CMS experiment in  $H \rightarrow \gamma\gamma$  decay mode [CMS:Hgg\_8TeV].

The Higgs cross section measurement helps to probe the structure, nature of the interaction of H boson with decaying particles. Differential fiducial cross sections are measured using the full Run 2 CMS data recorded at 13 TeV corresponding to several kinematic observables which are sensitive to the H boson production mechanisms: rapidity and transverse momentum of the H boson, associated jet multiplicity and transverse momentum of leading jet. These differential distributions are sensitive to production mechanism of gluon fusion, colliding protons' PDFs, theoretical modeling of hard quarks radiations and relative contributions of different production mechanisms of H boson.

The fragmentation of the dissertation is systematized as follows. An overview of the SM of particle physics, production and decays of H boson are described in Chapter 2. Overview of the LHC and its CMS experiment are presented in Chapter 3. The calibrations and reconstructions of the physics objects within the CMS detector are described in Chapter 4. Chapter 5 presents the procedure of measurements of inclusive and differential fiducial cross sections of H boson production at  $\sqrt{s} = 13$  TeV. Finally, the conclusions and outlook of the dissertation are summarized in Chapter 6.

# Chapter 2

## Theoretical Background

### 2.1 The Standard Model

The main goal of particle physics is to fully understand the nature of the constituents of the matter and the fundamental forces through which they interact. There are many theories purposed to understand the physics laws at particle level but the Standard Model(SM) is the one which stood firm against high precision tests [Reference1]. It is a collaborative theory of fundamental world, developed in 20th century which can express the fundamental particles (bosons and fermions) and the underlying interactions with the concept of mediating gauge bosons. The last unverified part of the SM, the Higgs boson, was discovered in 2012 at LHC of European Organization for Nuclear Research(CERN) [Reference20, Reference21]. Despite of incredibly successful theory of particle world, the SM does fail to fully describe the known universe.

### 2.2 Fundamental Particles and underlying Interactions

#### 2.2.1 Fundamental Particles

Particle with unknown structure is called fundamental particle. Fundamental particles can be categorized into fermions (characterized by Fermi-Dirac statistics) and bosons (Bose-Einstein statistics). The spin-statistics theorem classifies the particles with integral spin as bosons, while particles with half-integral spin as fermions. Fermions includes all leptons, quarks and baryons (composite particles three quarks). Ordinary matter is composed of leptons and quarks which are categorized in three “generations” as illustrated in Table 2.1. First generation includes an “up” type quark which carries electric charge  $+\frac{2}{3}$  and a “down” type quark carrying electric charge  $-\frac{1}{3}$ . Quarks also carry color charge (red, green, or blue) which is responsible for the strong interaction. The types of the quarks in order of increasing mass are “up”, “down”, “strange”, “charm”, “bottom” and “top”. The whole visible world is only made of light type of quarks, “up” and “down”. The heavy quarks can be produced only in high energy collisions (cosmic rays, particle accelerators) and quickly decays into “up” and “down” quarks.

Quarks interact through all three Standard Model forces, weak, strong, and electromagnetic. According to the Quantum Chromodynamics (QCD), two quarks

(quark antiquark pair) or three quarks can combine as mesons ( $q_1\bar{q}_2$ ) or baryons ( $q_1q_2q_3$ ) respectively to make color-neutral hadrons. Out of hadrons, protons and neutrons are most widely recognized making nuclei of matter which are formed of  $uud$  and  $udd$  quarks respectively.

| Fermions       | 1 <sup>st</sup> Gen.      | 2 <sup>nd</sup> Gen.        | 3 <sup>rd</sup> Gen.         | Charge         | Interactions          |
|----------------|---------------------------|-----------------------------|------------------------------|----------------|-----------------------|
| <b>Quarks</b>  | <i>up</i>                 | <i>charm</i>                | <i>top</i>                   | $+\frac{2}{3}$ | all                   |
|                | <i>down</i>               | <i>strange</i>              | <i>bottom</i>                | $-\frac{1}{3}$ |                       |
| <b>Leptons</b> | <i>e</i>                  | <i><math>\mu</math></i>     | <i><math>\tau</math></i>     | $-1$           | weak, electromagnetic |
|                | <i><math>\nu_e</math></i> | <i><math>\nu_\mu</math></i> | <i><math>\nu_\tau</math></i> | $0$            | weak                  |

TABLE 2.1: Generations, charge and interaction types of quarks and leptons.

Leptons also have three generations like quarks. Leptons only interact via electromagnetic and weak interaction, since they have no color charge. In fact, quarks and gluons are only particles carrying the color charge. Lepton generations each a charged lepton and a neutrino associated. Neutrinos are electrically neutral and carry no color charges, so interact through weak force only. Charged leptons masses are listed in order of increasing mass  $m_e = 0.511$  MeV,  $m_\mu = 105.7$  MeV and  $m_\tau = 1.78$  GeV. Charged leptons carry  $-1$  and their antiparticle carry  $+1$  electric charge. In addition, all leptons lepton flavor quantum number: all leptons have assigned  $+1$  while antileptons have assigned  $-1$ .

Carriers of fundamental forces are bosons, more details would be described in Section 2.2.2. Electromagnetic force is mediated by photon which is massless, neutral and of spin-1. On the other hand, weak force is carried by  $W^\pm$  (80.4 GeV) and  $Z$  (91.2 GeV) bosons having electric charges  $\pm 1$  and  $0$  respectively. At last, strong forces is carried by massless, neutral spin-1 gluons. Gluon is the only gauge boson which carries color and anti-color charge, thus it can interact with other gluons.

TABLE 2.2: Gauge bosons and their masses, coupling constants and interaction ranges for the three forces of the SM

| Force             | Weak                                              | Electromagnetic          | Strong                 |
|-------------------|---------------------------------------------------|--------------------------|------------------------|
| Mediator boson    | $W^\pm$ and $Z$                                   | photon ( $\gamma$ )      | Gluon (g)              |
| Mass(GeV)         | 80.4, 91.2 GeV                                    | 0                        | 0                      |
| Coupling Constant | $G_F \approx 1.2 \times 10^{-5} \text{ GeV}^{-2}$ | $\alpha = \frac{1}{137}$ | $\alpha_s \approx 0.1$ |
| Interaction Range | $10^{-16} \text{ cm}$                             | $\infty$                 | $10^{-13} \text{ cm}$  |

## 2.2.2 Fundamental Interactions

The three fundamental interactions (electromagnetic, strong and weak) are explained three dedicated theoretical descriptions Quantum Electrodynamics (QED),



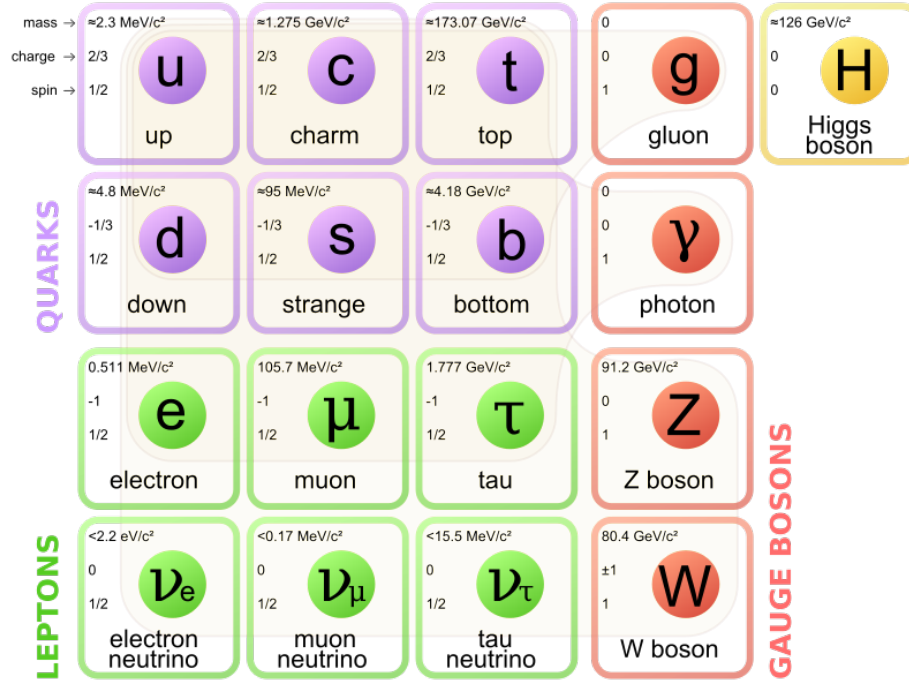


FIGURE 2.1: The elementary particles given in three generations of matter, the gauge bosons in fourth column and the Higgs boson in the fifth.

Quantum Chromodynamics (QCD) and Quantum Flavordynamics (QFD) which are encoded in SM. Electromagnetic and weak interactions were then unified into electroweak (EWK) interaction with significant contributions by Abdus Salam, Sheldon Glashow and Steven Weinberg. The mathematical description of interaction of charged particles are described by QED. Photon of QED is the force carrier of electromagnetic force. QED is the first theory which explains the interaction between light and matter that patches up special relativity and quantum mechanics. Figure 2.2 shows an example of QED, electron-positron annihilation. Annihilation is the process of destruction of particle and anti particle pair to any set of particles obeying conservation of quantum numbers, energy and momentum. QED is perturbative theory based on fundamentals of quantum mechanics established by three distinguished physicists W. Heisenberg, W. Pauli and P. A. M. Dirac during 1920s. Later on, Freeman J. Dyson, Richard P. Feynman and Julian S. Schwinger showed that charged particle can possibly interact through a sequence of processes with growing complexity, and all of these processes could be described by Feynman developed graphical technique. QED is abelian gauge theory with  $U(1)$  being its symmetry group.

Weak interaction, one of the four fundamental forces, was first revealed by the understanding of beta decay in 1933 by Enrico Fermi called Fermi's interaction [Referencea]. Weak force is mediated by  $W^\pm$  and  $Z$  bosons. In terms of a field theory, it is represented by gauge theory with symmetry group  $SU(2)$ , having three gauge bosons ( $W^+$ ,  $W^-$ ,  $Z$ ). Weak force, rarely known from quantum

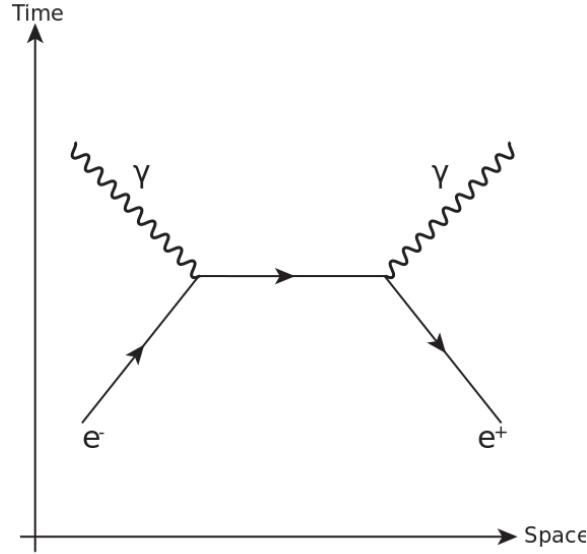


FIGURE 2.2: A Feynman diagram of annihilation of electron positron.

flavordynamics (QFD), is the only force which is proficient to change the flavor of quarks. This is very short range force because exchange bosons ( $W^\pm$ ,  $Z$ ) are massive. All fermions and the Higgs boson can interact via weak force. In 1970s, electromagnetic and weak forces were unified as electroweak force above unification energy (on the order of 100 GeV). The Physics Nobel Prize was awarded to Sheldon Glashow, Abdus Salam, and Steven Weinberg for their role in electroweak unification. In the field theory representation, Electroweak unification is settled under the gauge group  $SU(2)_L \otimes U(1)_\gamma$ . Experimental observation of existence of electroweak interaction was accomplished in two phases, the first was disclosure of existence of neutral current in scattering of neutrino in 1973 by Gargamelle collaboration and then the discovery of  $W$  and  $Z$  gauge bosons in proton-antiproton collisions by both the UA1 and the UA2 collaborations in 1983 [Reference2, Reference3, Reference4, Reference5].

The representation of SM is made by  $SU(3)_C \otimes SU(2)_L \otimes U(1)_\gamma$  gauge symmetry. Where  $U(1)_\gamma$  is representation of QED,  $SU(2)_L$  represents weak theory and  $SU(3)_C$  represents QCD. By the unification of electromagnetic and weak force represented by  $SU(2)_L \otimes U(1)_\gamma$  gauge symmetry [Reference6], Glashow, Salam and Weinberg was able to formulate the fields to account the  $W^\pm$ ,  $Z$  and  $\gamma$  (A) boson in footings of the original basis fields  $A'$  and  $B$  as expressed in equations 2.1, 2.2 and 2.3. The spontaneous breaking of  $SU(2)_L \otimes U(1)_\gamma$  gauge symmetry is primary reason for the generation of masses of the particles and separate electroweak (EW) interaction from electromagnetic (EM) and weak interactions. At higher energies ( $> 100 \text{ GeV}$ ), both weak and EM forces acts as similar style. Further details would be described in 2.3 section.

$$W_\mu^\pm = \sqrt{\frac{1}{2}}(A_\mu^1 \pm iA_\mu^2) \quad M_W = \frac{1}{2}gv \quad (2.1)$$

$$Z_\mu^\pm = \sqrt{\frac{1}{g^2 + g'^2}}(gA_\mu^3 - g'B_\mu) \quad M_Z = \frac{1}{2}v\sqrt{g^2 + g'^2} \quad (2.2)$$

$$A_\gamma = \sqrt{\frac{1}{g^2 + g'^2}}(gA_\mu^3 - g'B_\mu) \quad M_A = 0. \quad (2.3)$$

One of the the four fundamental forces is the strong force which is responsible for strong nuclear force which bounds the nucleus of an atom. As the name speaks for itself, strong force is strongest among four fundamental forces, although it act at a distance of  $10^{-15}$  meter (femtometer) or less. Strong force is described by QCD and only exist between quarks and gluons which carry non-vanishing color charges. In the field theory representation of the SM, the strong force is represented by the symmetry group of  $SU(3)_C$ . Here, C represents QCD color, which must be conserved in all such interactions. The strong interaction is mediated by gluons and contrary to the other fundamental forces, strong force does not diminish with gain in distance between quarks know as color confinement. As an implication of this property, no signature of elementary quark or gluon particle are observed in High Energy Physics (HEP) colliders, instead they bound with each other to form streams of hadrons by phenomenon known as hadronization called jets.

## 2.3 The Brout-Englert-Higgs Mechanism (BEH)

Gauge fields must acquire mass in order to correctly describe weak interaction and to be consistent with experimental findings. Before electroweak symmetry breaking (EWSB), all particles were supposed not to have masses, however mass term in electroweak Lagrangian for a gauge boson must be introduced to guarantee the renormalization of theory that would violate gauge invariance. Which hints that some basic thing is missing in the theory apart from fermions and gauge boson fundamental fields, which can maintain the local gauge invariance. The Higgs mechanism was postulated as a part of SM, which introduces the existence of a new scalar Higgs field. This field transformation ensures the renormalizability of the theory and gauge invariance. By this formulation the concept of breaking of symmetry and origin of masses of the fermions and bosons ( $W^\pm$ ,  $Z$  boson) are introduced [Reference7, Reference8, Reference9, Reference10, Reference11, Reference12]. The Higgs mechanism introduces a doublet (under isospin) of complex scalar field

$$\phi = \begin{pmatrix} \phi^\dagger \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^1 + i\phi^2 \\ \phi^3 + i\phi^4 \end{pmatrix} \quad (2.4)$$

which can be substituted in electroweak lagrangian given below

$$\mathcal{L}_{EWSB} = (D_\mu \phi)^\dagger (D_\mu \phi) + V(\phi^\dagger \phi) \quad (2.5)$$

here  $D_\mu = \delta_\mu -igt_a W_\mu^a + \frac{i}{2}g'Y B_\mu$  represents covariant derivative and the lagrangian described in Equation 2.5 is gauge invariant under the  $SU(2)_L \otimes U(1)_\gamma$  transformation. The kinetic energy term in the lagrangian is written in terms of the covariant

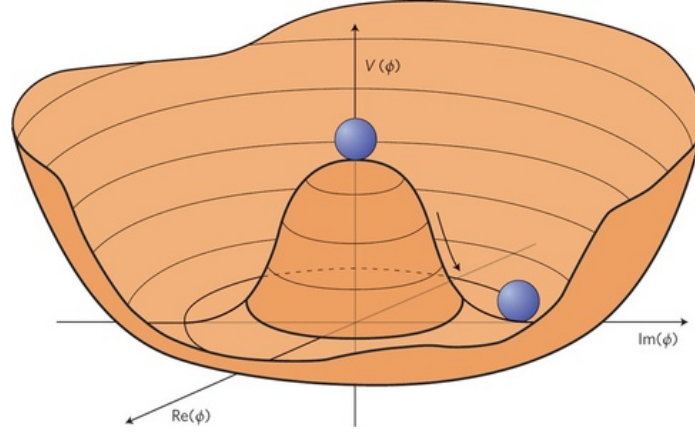


FIGURE 2.3: The Higgs potential as in Eq.2.6

derivatives whereas the potential only depends upon the product  $\phi^\dagger \phi$ .  $\phi$  is the scalar field and can be codified in terms of quantum numbers which are isospin ( $I_3$ ), hypercharge (Y) and electric charge (Q):

|                                                        | $I_3$                                                       | Y                                      | Q                                      |
|--------------------------------------------------------|-------------------------------------------------------------|----------------------------------------|----------------------------------------|
| $\begin{pmatrix} \phi^\dagger \\ \phi^0 \end{pmatrix}$ | $\begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$ | $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ | $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ |

while the potential term in Equation 2.5 can also be expressed as

$$V(\phi^\dagger \phi) = -\mu^2 \phi^\dagger \phi - \lambda (\phi^\dagger \phi)^2 \quad (2.6)$$

here  $\mu$  is mass term and the parameter  $\lambda$  represents the strength of the interaction. In the scenario,  $\mu > 0$  and  $\lambda > 0$  and the potential shapes becomes very interesting, which is essential for Higgs mechanism shown in Figure 2.3. There occurs a minima for

$$\phi^\dagger \phi = \frac{1}{2}(\phi_1^\dagger + \phi_2^\dagger + \phi_3^\dagger + \phi_4^\dagger) = \frac{-\mu^2}{2\lambda} \equiv \frac{V^2}{2} \quad (2.7)$$

but on the circle of points where the magnitude of  $z$  is  $\phi$ . This minimum does not exist at a single value of  $\phi$ , but at circular non-zero values. The  $(\phi^\dagger \phi^0)$  selection that affiliated with the ground state is arbitrary and in  $(\phi^\dagger \phi^0)$  plane the selected point is not invariant under rotations which is called spontaneous symmetry breaking. To settle the ground state on the  $\phi^0$  axis, the vacuum expectation value (VEV) of field  $\phi$  is

$$\langle \phi \rangle = \frac{1}{\sqrt{2}} = \begin{pmatrix} 0 \\ V \end{pmatrix}, \quad V^2 = -\frac{\mu^2}{\lambda}. \quad (2.8)$$

The field  $\phi$  given below, is derived by rewriting it in some generic gauge, on footings of its VEV:

$$\phi = \frac{1}{\sqrt{2}} e^{i\phi^a t_a} \begin{pmatrix} 0 \\ H + \nu \end{pmatrix}, \quad a = 1, 2, 3 \quad (2.9)$$

here  $\phi_a$  and  $\phi_4 = H + \nu$  represents massless fields called Goldstone fields. New degrees of freedom is introduced along with six degrees because of polarizations vector bosons  $W^\pm$  and  $Z$ , as they are massless and scalar. A unitary gauge transformation in the field is fixed by

$$\phi' = e^{-\frac{i}{\nu}\phi^a t_a} \phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ H + \nu \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \phi^4 \end{pmatrix}$$

so here we have zero expectation value for the Higgs field.

Now the lagrangian in Eq. 2.5 can be rewritten and disintegrated into three components

$$\mathcal{L}_{EWSB} = \mathcal{L}_H + \mathcal{L}_{HW} + \mathcal{L}_{HZ} \quad (2.10)$$

where three terms is given, assuming approximation  $V = \mu^2 H^2 + \cos t$  and ignoring higher order terms:

$$\begin{aligned} \mathcal{L}_H &= \frac{1}{2} \delta_a H \delta^a H + \mu^2 H^2 \\ \mathcal{L}_{HW} &= \frac{1}{4} \nu^2 g^2 W_a W^{\dagger a} + \frac{1}{2} H W_a W^{\dagger a} \end{aligned} \quad (2.11)$$

$$\begin{aligned} &= m_W^2 W_a W^{\dagger a} + g_{HW} W_a W^{\dagger a} \\ \mathcal{L}_{HZ} &= \frac{1}{8} \nu^2 (g^2 + g'^2) Z_a Z^a + \frac{1}{4} \nu (g^2 + g'^2) H Z_a Z^a \\ &= \frac{1}{2} m_Z^2 Z_a Z^a + \frac{1}{2} g_{HZ} H Z_a Z^a. \end{aligned} \quad (2.12)$$

For  $W^\pm$  and  $Z$  bosons, mass terms are included in Eqs. 2.11 and 2.12. This adds up three additional degree of freedom; however, three of the total four Goldstone bosons have also been vanished from the Lagrangian  $\mathcal{L}_{EWSB}$  not affecting overall number of degrees of freedom. A new scalar field has been introduced referred as Higgs field, introduced in the theory. The boson associated with Higgs field is called Higgs boson. The mass of the Higgs boson is linked with a free parameter in the theory  $\lambda$  that depends upon the self-interaction of Higgs field.

Briefly, the BEH mechanism ensures the renormalizability of theory and explain existence of the massive weak gauge bosons, despite of the fact that the fundamental gauge boson in theory still stays to be massless. In fact, the symmetry still exist in equation so using “hidden” is preferred over broken. According to BEH mechanism, fermions also acquire masses via their interaction with Higgs field. The invariance of the minimum for the Higgs field for  $U(1)_\gamma$  group shows the electromagnetic symmetry is not spontaneously broken that keeps photon massless.

### 2.3.1 Vector Boson Masses and Couplings

Eqs. 2.11 and 2.12 shows the masses of  $W^\pm$  and  $Z$  bosons depends upon a parameter  $\nu$  which is vacuum expectation value that defines a mass scale property of

EWSB and to electroweak coupling constants:

$$\begin{cases} m_W = \frac{1}{2}\nu g \\ m_Z = \frac{1}{2}\sqrt{g^2 + g'^2} \end{cases} \rightarrow \frac{m_W}{m_Z} = \frac{g}{\sqrt{g^2 + g'^2}} = \cos(\theta_W). \quad (2.13)$$

Also, Higgs boson couplings to vector bosons can also be obtained by Eqs. 2.11 and 2.12

$$g_{HW} = \frac{1}{2}\nu g^2 = \frac{2}{\nu}m_W^2 \quad (2.14)$$

$$g_{HZ} = \frac{1}{2}\nu(g^2 + g'^2) = \frac{2}{\nu}m_Z^2. \quad (2.15)$$

Using above Eqs. 2.14 and 2.14 helps to get the ratio of Higgs decay to  $W^\pm$  and  $Z$  bosons

$$\frac{\mathcal{B}(H \rightarrow W^+W^-)}{\mathcal{B}(H \rightarrow ZZ)} = \left( \frac{g_{HW}}{\frac{1}{2}g_{HZ}} \right)^2 = 4 \left( \frac{m_W^2}{m_Z^2} \right)^4 \approx 2.4. \quad (2.16)$$

At last, the characteristic mass scale for EWSB may be obtained by the relation between the Fermi constant  $G_F$  and the  $\nu$  parameter:

$$\nu = \left( \frac{1}{\sqrt{2}G_F} \right) \approx 246 \text{ GeV}. \quad (2.17)$$

### 2.3.2 Fermions Masses and Couplings

Fermions couples to Higgs field to acquire mass by Higgs mechanics. This can be achieved by introducing a  $SU(2)_L \times U(1)_\gamma$  invariant term (called yukawa term) in SM lagrangian that describes the interactions between the fermion fields and the Higgs. As  $\phi$  is an iso-doublet and fermions can be disintegrated up into right-handed and left-handed singlet, the yukawa term for each fermion will have expression given below for leptons:

$$\mathcal{L}_\ell = -G_\ell \bar{l}_\ell \phi l_R + \bar{\ell}_R \phi^\dagger l_\ell. \quad (2.18)$$

As the first component of  $\phi$  is zero, so a mass term will only emerge for the second component of  $l_\ell$ : this exactly repeat the fact that neutrino is massless.

$$\mathcal{L}_\ell = -\frac{G_{H\ell}}{\sqrt{2}} \nu \bar{\ell} \ell - \frac{G_{H\ell}}{\sqrt{2}} H \bar{\ell} \ell. \quad (2.19)$$

For quarks, the down type quarks (d, s, b) behave in similar way as leptons while up type quarks (u, c, t) must couple to charge conjugate of  $\phi$

$$\phi_c = -i\tau_2 \phi^* = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^3 + i\phi^4 \\ -\phi^1 + i\phi^2 \end{pmatrix} \quad (2.20)$$

in unitary gauge that becomes

$$\phi_c = \frac{1}{\sqrt{2}} \begin{pmatrix} \eta + \nu \\ 0 \end{pmatrix}. \quad (2.21)$$

Therefore, the yukawa Lagrangian can be expressed as

$$\mathcal{L} = -G_{H\ell} \bar{L}_L \phi \ell_R - G_{Hd} \bar{Q}_L \phi d_R - G_{Hu} \bar{Q}_L \phi^c u_R + h.c.. \quad (2.22)$$

The couplings constant and mass of a fermion can be derived from Eq. 2.19, written as

$$m_f = \frac{G_{Hf}}{\sqrt{2}} \nu \quad (2.23)$$

$$g_{Hf} = \sqrt{G_{Hf}} \sqrt{2} = \frac{m_f}{\nu}. \quad (2.24)$$

The fermions mass can not be predicted by the theory as  $G_{Hf}$  is free parameters. The interaction of Higgs field with fermions transforms right-handed chirality in a left-handed chirality and vice versa in a way that object acquire mass. The strength of the interaction of Higgs fields with particles is reflected by its mass.

This evaluation is rather interpretative, as it not accumulating the fact that there are three families of quarks and leptons in reality. The different between weak mass eigenstates and the physical mass eigenstates makes real analysis complicated. Transformation of the weak mass basis to physical basis is essential to deal with the physical particles. This is achieved by the usual unitary transformation matrices, Cabibbo-Kobayashi-Maskawa (CKM) and Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrices, for quark and lepton sectors respectively.

## 2.4 Production and Decay of the Higgs Boson

After electroweak unification, spontaneous symmetry breaking and Higgs mechanism solves all the puzzles with the assumption of existence of the Higgs boson. Higgs boson was considered as the last missing patch of the Standard Model. Higgs boson has been searched at LHC within full mass range allowed given unitary and above LEP constraints, i.e from 114 GeV to 1 TeV. In this Section, Higgs boson production and decay modes would be discussed. This will allow us to identify the most promising channels in the perspective of Higgs boson discovery and its properties measurements. Theory does not predict the Higgs mass and coupling of the Higgs with bosons are directly proportional to the square of their masses as shown in Eqs. 2.14 and 2.15. In case of fermions, their couplings with the Higgs is directly proportional to the mass of the fermion as in Eq. 2.24. For this reason, the most promising channels for Higgs searches are its decays to  $W^+W^-$ ,  $ZZ$  and third generation of leptons. Also, photon being massless particle does not couple to Higgs directly but through loop contributions dominated by top quark and  $W^\pm$  boson.



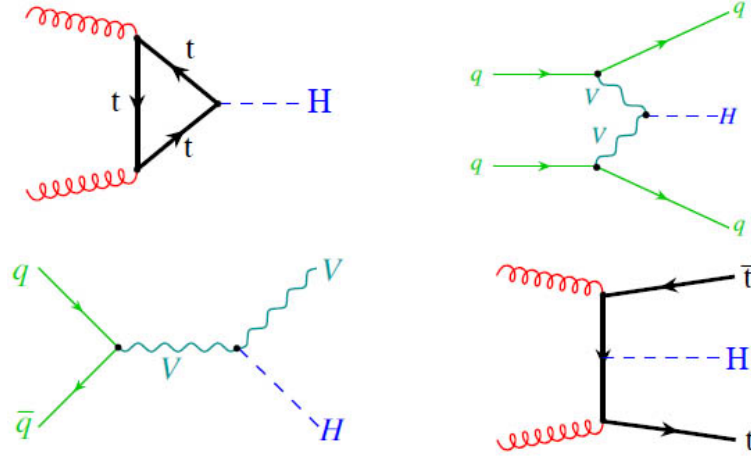


FIGURE 2.4: Higgs production mechanisms at tree level in proton-proton collisions: gluon-gluon fusion (top left); VV fusion (top right); W and Z associated production (bottom left);  $t\bar{t}$  associated production (bottom right).

### 2.4.1 Higgs Production

The dominant production modes of the Higgs boson in pp collisions at LHC are shown in Figure 2.4, corresponding production cross sections as a function of the Higgs mass at center-of-mass energies  $\sqrt{s} = 7, 8$  and 13 TeV are shown in Figure 2.5. A considerable increase in the Higgs production cross section is found as moves towards higher center-of-mass energies shown in Figure 2.6 [Referenceb] for Higgs mass 125 GeV.

#### Gluon-gluon Fusion

This is most dominant Higgs production mode at LHC in whole mass range and center-of-mass energies range is gluon-gluon fusion ( $gg \rightarrow H$ ), illustrated in Figure 2.4 (top left) [Referenceb]. As LHC and tevatron are hadron colliders, it highly probable that two gluons responsible for binding of hadron collide. The gluons couples to Higgs through the loop of virtual quarks, as coupling to Higgs is directly proportional to the mass of fermions, top quark loop is most dominant. The  $ggH$  cross section for Standard Model Higgs at  $m_H = 125$  GeV at center-of-mass energies of 7 TeV, 8 TeV and 13 TeV are approximately 15, 19, 44 pb respectively. As we move towards higher energies the production cross section increases significantly, i.e 27% from 7 TeV to 8 TeV and 227% from 8 TeV to 13 TeV at  $m_H = 125$  GeV. The cross section numbers in Table 2.3 are computed at NNLO QCD and NLO EW accuracies.



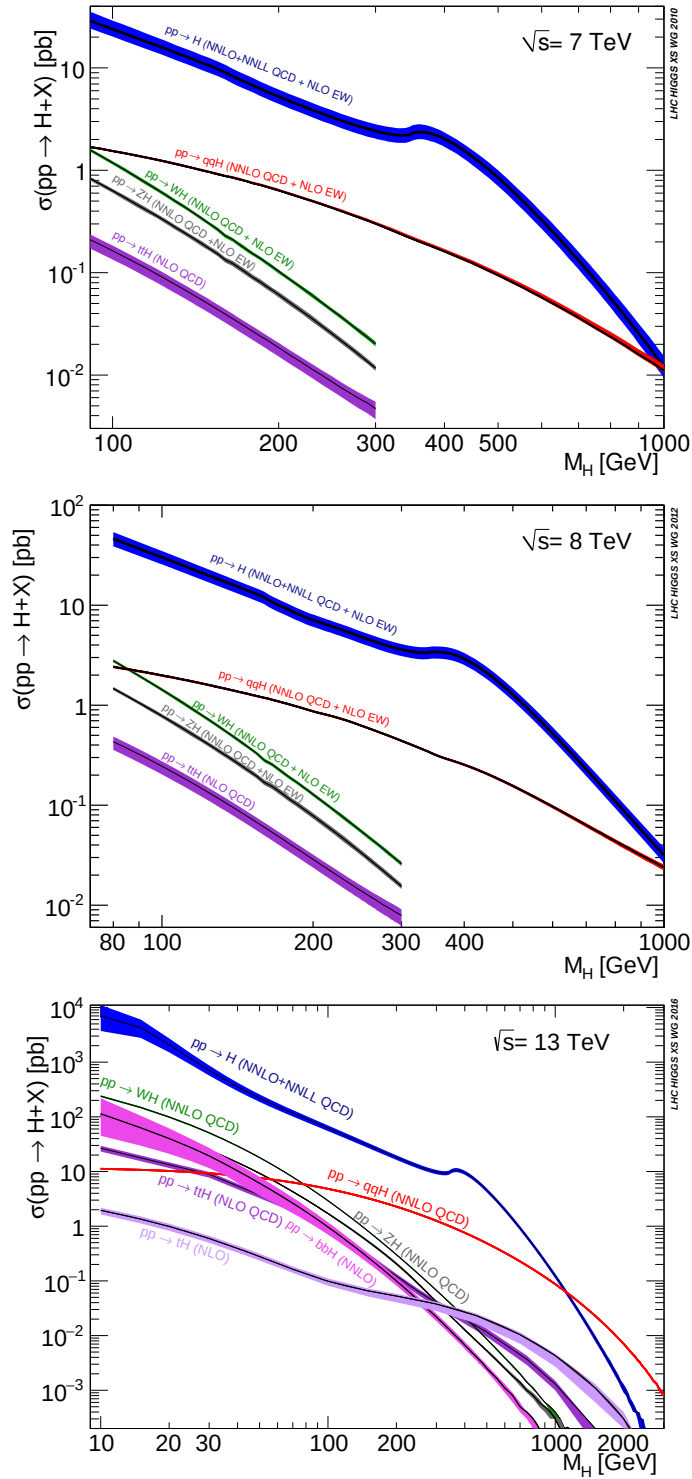


FIGURE 2.5: Standard Model Higgs boson production cross sections at 7, 8 and 13 TeV as function of mass of Higgs boson.

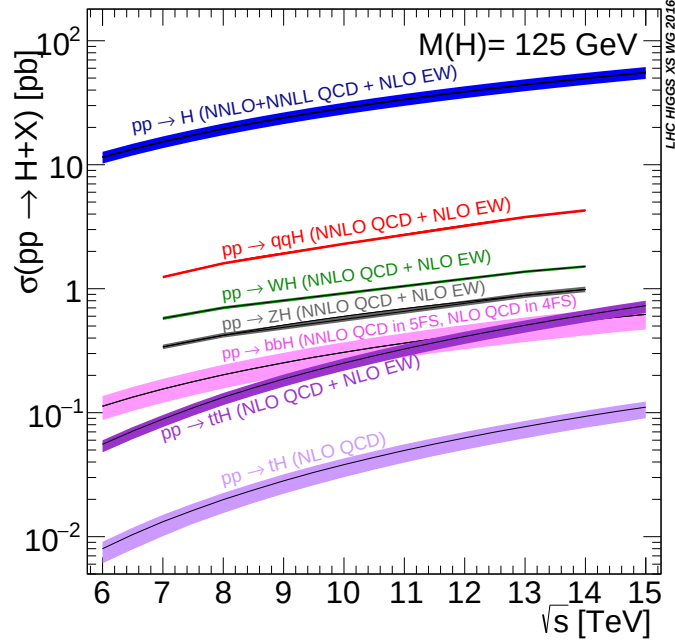


FIGURE 2.6: Higgs boson production cross sections as a function of center-of-mass energies at  $m_H = 125$  GeV

TABLE 2.3: Higgs boson production cross sections in pb for different Higgs masses and center-of-mass energies by ggH and VBF productions mechanism at LHC.

| $m_H(\text{GeV})$ | ggH   |       |        | VBF   |       |        |
|-------------------|-------|-------|--------|-------|-------|--------|
|                   | 7 TeV | 8 TeV | 13 TeV | 7 TeV | 8 TeV | 13 TeV |
| 120               | 16.61 | 21.07 | 47.38  | 1.30  | 1.68  | 3.9    |
| 125               | 15.31 | 19.47 | 44.14  | 1.24  | 1.60  | 3.78   |
| 130               | 14.15 | 18.03 | 41.23  | 1.19  | 1.53  | 3.64   |

### Vector Boson Fusion

Vector boson fusion (VBF) is the second most important Higgs production mode at LHC and LEP as shown in Figure 2.4 (top right). VBF Higgs production cross section is roughly 10 times lower than ggH at  $m_H = 125$  GeV, and becomes comparable at  $m_H = 1$  TeV. The VBF cross section for Standard Model Higgs at  $m_H = 125$  GeV at center-of-mass energies of 7 TeV, 8 TeV and 13 TeV are approximately 1.2, 1.6, 3.8 pb respectively. The cross section numbers in the Table 2.3 are computed at NNLL QCD and NLO EW precision [Referenceb]. In VBF production mode, two  $W^\pm$  or  $Z$  boson fusion create Higgs that have been emitted by a (anti)fermion pair collision. The VBF experimental signal is very clean and can be well distinguished from ggH with two high energy jets in forward region of the detector. These jets signature helps to increase signal to background ratio.

TABLE 2.4: Higgs boson production cross section in pb for different Higgs masses and center-of-mass energies by associative productions mechanism at LHC.

| $m_H(\text{GeV})$ | WH    |       |        | ZH    |       |        | ttH   |       |        |
|-------------------|-------|-------|--------|-------|-------|--------|-------|-------|--------|
|                   | 7 TeV | 8 TeV | 13 TeV | 7 TeV | 8 TeV | 13 TeV | 7 TeV | 8 TeV | 13 TeV |
| 120               | 0.66  | 0.81  | 1.57   | 0.39  | 0.48  | 0.99   | 0.11  | 0.15  | 0.57   |
| 125               | 0.58  | 0.70  | 1.37   | 0.34  | 0.42  | 0.88   | 0.09  | 0.13  | 0.51   |
| 130               | 0.51  | 0.62  | 1.21   | 0.30  | 0.37  | 0.79   | 0.08  | 0.12  | 0.45   |

### W and Z Associated Production

W and Z associated production also called Higgs Strahlung is defined by Higgs production in association with  $W^\pm$  or  $Z$  boson as shown in Figure 2.4 (bottom left). In this production mechanism, Higgs is radiated by high energetic  $W^\pm$  or  $Z$  bosons that is produced by fusion of two fermions. This production mechanism was leading production mode at LEP and sub-leading at Tevatron. At LEP it becomes very probable to produce  $W^\pm$  or  $Z$  boson with electron positron collisions. In case of hadron collider, Tevatron makes quark antiquark collisions more likely as compared to LHC. At LHC this production mechanism is at third number. The WH and ZH cross section for Standard Model Higgs at center-of-mass energies of 7 TeV, 8 TeV and 13 TeV are listed in Table 2.4 which are computed at NNLL QCD and NLO EW precision [Referenceb].

### Top Fusion

The least dominant process of Higgs production is illustrated in Figure 2.4 (bottom right). In this process two gluons collide producing pair of top antitop which produces Higgs by fusion in association  $t\bar{t}$ . The ttH cross section for Standard Model Higgs at center of mass energies of 7 TeV, 8 TeV and 13 TeV are listed in Table 2.4 which are computed at NLO QCD and NLO EW precision [Referenceb].

## 2.4.2 Higgs Decay

As Higgs boson is electroweak particle, so it wants to decay quickly. The Higgs boson decay likelihood depends on a variety of aspects, i.e mass of the Higgs boson, interaction strength etc, most of these factors are controlled by SM except the Higgs boson mass itself. The life time of SM Higgs boson with mass of 126 GeV is about  $1.6 \times 10^{-22}$  s. Higgs boson can decay to all SM particles with mass but it likes to decay most heavier particles available. Higgs boson decay branching ratio to a variety of particles are illustrated in Figure 2.7 [Referenceb].

The most dominant decay channel of the Higgs boson  $H \rightarrow b\bar{b}$  at  $m_H \approx 125$  GeV. But the QCD background in this channel is of many order higher than signal which makes it less sensitive channel. However, the  $b\bar{b}$  channel has been exploited in the boosted regime, in association with a vector boson decaying leptonically, by CMS [Reference13], by ATLAS [Reference14] and by CDF and D0 [Reference15]. B quarks are also difficult to reconstruct in a detector. When they decay in the

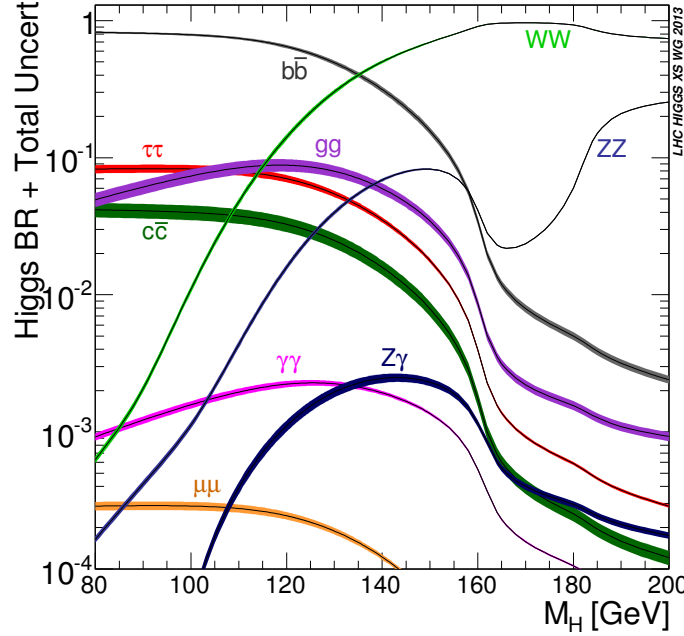


FIGURE 2.7: Branching ratios for different Higgs boson decay channels as a function of the Higgs mass.

detector, they fragment into jets of many hadrons. These jets are difficult to reconstruct perfectly. Because of this, it is necessary to exploit other channels in this region.  $H \rightarrow \gamma\gamma$  is one most favorable decay channel due to signature with two high energy photons with excellent mass resolution.

The main backgrounds in this channel are  $q\bar{q} \rightarrow \gamma\gamma$ ,  $gg \rightarrow \gamma\gamma$  and instrumental background in which one or two jets are identified as photons. As the Higgs mass at 125 GeV, Higgs boson decay to diboson( $WW$ ,  $ZZ$ ) boson are also available, where one or both boson can be off-shell. In case of  $WW$  decay, its impossible to completely reconstruct  $WW$  boson in detector as it decays  $H \rightarrow WW \rightarrow \ell\nu\ell\nu$  have two neutrinos and  $H \rightarrow WW \rightarrow \ell\nu jj$  have one neutrino and two jets. These objects pulls mass distribution to be wide and signal to background ratio to be low, which makes  $WW$  decay channel less sensitive. While  $H \rightarrow ZZ \rightarrow \ell\ell\ell\ell$  decay is labeled as golden channel in spite of low branching ratio. The clean signature of at least four leptons makes this channel almost background free. These leptons are high energetic leptons which can reconstructed with excellence in the detector. The branching ratio of the Higgs boson in intermediate range is illustrated in Figure 2.8 [Referenceb].

## 2.5 Experimental Observations

The electroweak unification theory lead to series of impressive experimental discoveries. In 1973, the evidence of neutral current through inelastic scattering was

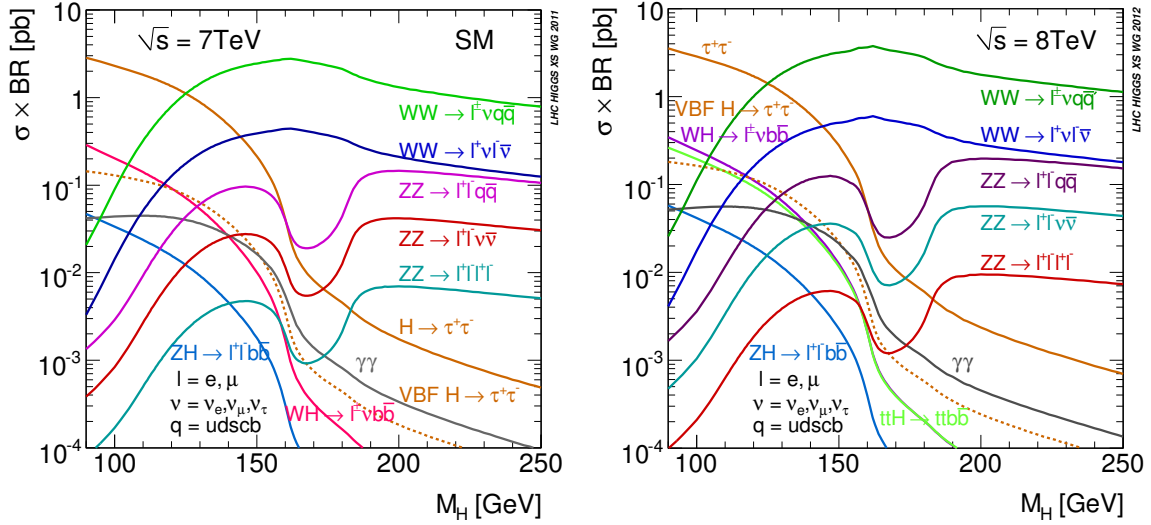


FIGURE 2.8: Higgs cross sections times branching ratios 7 TeV (left) and 8 TeV (right).

found by Gargemelle bubble chamber collaboration at CERN [Reference16]. This evidence provide starting platform to the Standard Model of electroweak interaction.

Super Proton Synchrotron proton-antiproton collider at CERN played its role to next breakthrough that came a decade later. The existence of  $W$ ,  $Z$  boson announced by UA1 and UA2 collaboration through the channels  $Z \rightarrow e^+e^-$ ,  $Z \rightarrow \mu^+\mu^-$ ,  $W \rightarrow e\nu$  [Reference2, Reference3, Reference4, Reference5]. This was a first observation of  $W$ ,  $Z$  boson that mediates weak force. Carlo Rubbia and Simon van der Meer was awarded with Nobel Prize for their contribution in 1984.

These important experimental discoveries and growing knowledge on weak interaction convinced to construct the Large Electron Positron collider. at CERN, for deeper look into  $W$ ,  $Z$  further decays, their precise mass measurement [Reference17, Reference18]. LEP was equipped with four detectors around the ring; ALEPH (Apparatus for LEP Physics), DELPHI (Detector with Lepton and Hadron Identification), OPAL (Omni-Purpose Apparatus for LEP) and L3 (Letter of Intent 3). These experiments reported many precise results of the Standard Model, specially measuring the mass of  $W$  and  $Z$  boson and confirming three neutrino species consistent with Standard Model [Reference18].

Around 2000, Large Hadron Collider [Reference19] at CERN construction started to look for last missing patch of electroweak unification, the Higgs boson. LHC is equipped with four major detectors; ATLAS (A Toroidal LHC Apparatus), CMS (Compact Muon Solenoid), ALICE (A Large Ion Collider Experiment), LHCb (Large Hadron Collider beauty). LHC recorded data of proton-proton collisions at center-of-mass energies 7 and 8 TeV in 2011 and 2012. After a long search, the CMS and ATLAS collaboration, two general purpose experiment at LHC, announced the exist of the heavy scalar boson on 4 July 2012 [Reference20, Reference21]. Further test to measure of the properties of this scalar boson immediately begins, in March 2013, CERN announced the tentative confirmation of consistency of its properties

with SM [Reference22, Reference23, Reference24, Reference25].

## 2.6 Beyond Standard Model

Although SM has passed numerous experimental test with great precision, it is not considered to be theory of everything. There are many phenomenons which are not explained by SM i.e Hierarchy problem, existence of dark matter and dark energy, neutrinos masses etc. There are several theoretical extension posed in Standard Model such as Minimal Super Symmetrical Model(MSSM) and Next-to-Minimal Super Symmetric Standard Model (NMSSM) or completely new theoretical description, such as string theory, extra dimension. The simplest broadening of the SM Higgs sector is the the electroweak singlet (EWS) model [Reference26]. In this model, a heavy singlet is defined along with SM scalar Higgs field and then two CP-even Higgs bosons comes out as result of spontaneous symmetry breaking. Another generic extension to the SM Higgs is the 2 Higgs Doublet Model (2HDM), which introduces two scalar, complex doublet  $\phi_1$  and  $\phi_2$  [Reference27]. Spontaneous symmetry breaking results five Higgs boson, a neutral CP-odd one (A), two CP-even particles (h and H) and two charged ( $H^\pm$ ). Experiments are looking for any direct or indirect hints of new physics. Some of main aspects where SM is unable to explain are discussed below.

### 2.6.1 Hierarchy Problem

The larger discrepancy between the features of the gravity and the weak force is known as “Hierarchy Problem”. There is no convincing explanation of the problem that why the gravity force is  $10^{32}$  times weaker than weak force. The field theory breaks down because of existence of quantum gravity at plank scale ( $M_p \approx 10^{19}$  GeV). Writing the Standard Model Lagrangian for the Higgs mass will lead to following expression below

$$m_H^2 = m_{oH}^2 + \delta m_H^2 \quad (2.25)$$

here  $m_{oH}^2$  represents mass parameter exist in the Standard Model Lagrangian, and  $\Delta m_H^2$  represents the quantum radiative corrections to the mass that are predicted to be heavier. These corrections can be found as

$$\Delta m_H^2 \approx \frac{\lambda^2}{16\pi^2} \Lambda^2 \quad (2.26)$$

here  $\lambda$  represents the Yukawa coupling constant, and  $\Lambda$  represents the energy at which the Standard Model is invalid, or about 1 TeV. This was also one of the reason to expect Higgs boson mass at order of few hundred GeV. The Super symmetry (SUSY) provide the solution of Hierarchy problem by predicting the existence of heavier super partners of all SM particles (bosons and fermions). Several direct and indirect searches has been conducted at LHC to hunt for SUSY, but no evidence has been found so far.

### 2.6.2 Dark Matter and Dark Energy

Latest observation from cosmological experiments shows that universe is composed of 4.9% ordinary matter, 26% dark matter and 70% dark energy. Standard Model does not provide any clue about existence of dark matter and dark energy.

### 2.6.3 Matter Antimatter Asymmetry

Today, if we look around in the universe, we see matter dominance over antimatter. At the time of Big-Bang matter and antimatter are supposed to be produced in equal quantities. This give rise to one of the greatest mysteries, where all antimatter has gone. Standard Model is unable to explain this imbalance between matter and antimatter.

### 2.6.4 Grand Unification and Beyond

One of main goal of the particle physics, is to unify all fundamental force at some energy scale to describe the theory of everything as Standard Model unify electromagnetic and weak force in to electroweak. But Standard Model does not provide any clue about unification of gravity along with other fundamental force generally referred as Grand Unification Theory (GUT). This implies that Standard Model itself is a patch of Grand Unification. The search for any new physics (beyond Standard Model) is important and justified that could improve our understanding about these unsolved mysteries.

## 2.7 Higgs Boson as a Probe to BSM

After the discovery of the Higgs boson, it became very natural and important to measure Higgs boson properties as precise as possible. Measurements of the Higgs boson properties provide a direct probe into SM Higgs sector perturbative QCD calculations at the TeV energy scale.

The couplings of the Higgs boson with fermions are predicted by SM through Yukawa interactions. Quarks and leptons acquire masses through Yukawa interactions with Higgs boson. While in SM the coupling strength of Higgs boson with any other particle is directly proportional to the mass of that particle. The masses of the leptons and quarks are well know by experiments, so it possible to estimate the decay rates of Higgs to fermions precisely. Experimental measurements can test these prediction which will provide a direct hint to BSM if any deviation is observed.

The cross sections measurement of the Higgs boson illustrate the possibility of the occurrence of that process. Any deviation from SM prediction in integrated or differential cross sections will guide to BSM effects. The observation and studies of the Higgs boson decays like gluon gluon fusion ( $gg \rightarrow H$ ), vector boson fusion (VBF), associated production with a vector boson (WH, ZH) and top quark pair ( $t\bar{t}H$ ) also shed light on the coupling of the Higgs with vector boson, top quark etc.





## Appendix

